Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super60**(In Coming) | **Jee-Main** | Date:04-09-2021

Time: 09.00Am to 12.00 | **PTM-04** | Max.Marks:300

Key Sheet

PHYSICS

1	2	2	3	3	1	4	3	5	1
6	1	7	3	8	2	9	3	10	2
11	2	12	2	13	1	14	2	15	3
16	2	17	3	18	3	19	3	20	3
21	4.00	22	2.00	23	3.00	24	3.00	25	3.00
26	4.00	27	1.50	28	3.50	29	2.00	30	0.00

CHEMISTRY

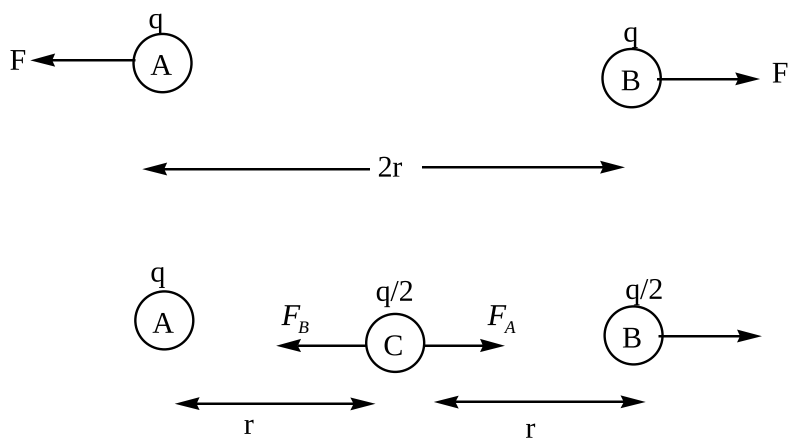
31	2	32	2	33	3	34	2	35	1
36	4	37	3	38	2	39	2	40	3
41	4	42	2	43	3	44	3	45	3
46	2	47	1	48	1	49	1	50	1
51	4	52	5	53	3	54	5	55	2
56	6	57	3	58	27	59	4	60	9

MATHEMATICS

61	3	62	4	63	3	64	2	65	2
66	2	67	4	68	4	69	2	70	1
71	3	72	2	73	4	74	2	75	4
76	2	77	3	78	4	79	2	80	2
81	3	82	4	83	2	84	7	85	1.5
86	0.75	87	12	88	0	89	1	90	4

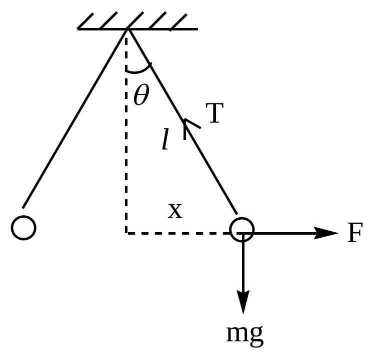
SOLUTIONS
PHYSICS

1.



Net force on C is $F_A - F_B = F$

2.



$\tan\theta = \frac{F}{mg}$

In air, $\tan\theta = \frac{1}{mg} \times \frac{1}{4\pi k \in_0} \frac{q^2}{[2x]^2}$ (1)

In medium, $\tan\theta = \frac{1}{mg\left[1-\frac{\rho_l}{\rho_b}\right]} \times \frac{1}{4\pi k \in_0 k} \frac{q^2}{(2x)^2}$ (2)

From (1) & (2) $\rho_b = 12 \quad gm/c.$c

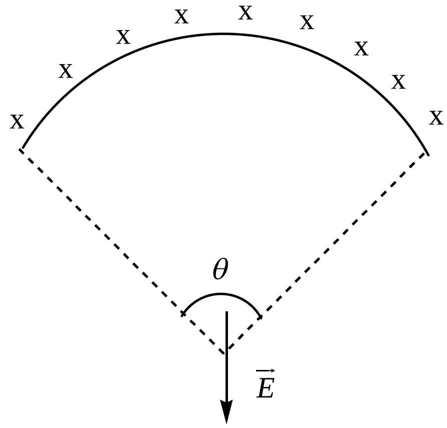
3. $\frac{dF}{dm} = 0$, then F becomes maximum,

$\frac{d}{dm}\left[\frac{Gm(M-m)}{r^2}\right] = 0$

$\Rightarrow \frac{m}{M} = \frac{1}{2}$

4. Electric field due to curved charged wire at its centre of curvatine

$E = \frac{\lambda \sin(\theta/2)}{2\pi \in_0 r}$



and due to straight charged wire at '0' field is zero

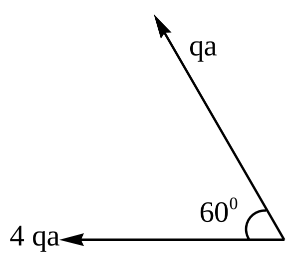
then $E_{net} = E_{(1m)} - E_{(2m)} + 0 + 0 = 9000\,N/C$

5. $\vec{E} = \frac{200}{\sqrt{2}}\left(\hat{i} + \hat{j}\right)$

$\overrightarrow{r_{PQ}} = \overrightarrow{r_P} = \overrightarrow{r_Q} = \hat{i} - 2\hat{j}$

$V_P - V_Q = \vec{E}.\overrightarrow{r_{PQ}} = \frac{200}{\sqrt{2}}volt$

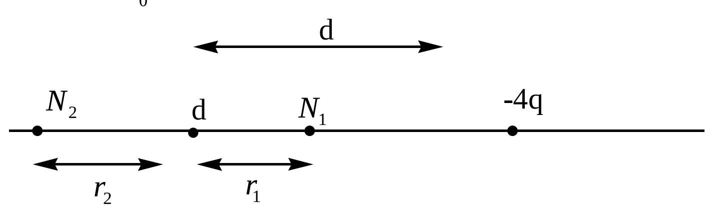
6.



$p = \sqrt{(qa)^2 + (4qa)^2 + 2(qa)(4qa)\cos 60^0}$

7. $\vec{E} = \overrightarrow{E_1} + \overrightarrow{E_2} = \frac{1}{4\pi\varepsilon_0}\frac{2p}{a^3}\hat{i} - \frac{1}{4\pi\varepsilon_0}\frac{p}{a^3}\hat{j}$

$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}\frac{\sqrt{5}.p}{a^3}$



8.

Two, on either side of smaller charge

at $N_1 \Rightarrow V_q + V_{-4q} = 0$

$\frac{1}{4\pi\in_0}\frac{q}{r_1} + \frac{1}{4\pi\in_0}\frac{(-4q)}{(d - r_1)} = 0$

Similarly at N_2

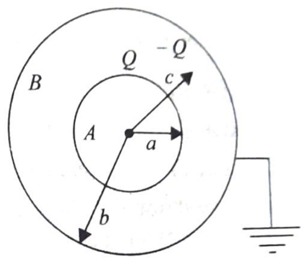
9. Potential is zero on equatorial line of an electric dipole.

10. Charge +Q will flow into earth

$$V_C = \frac{kQ}{r} - \frac{kQ}{b}$$

$$V_A = \frac{kQ}{a} - \frac{kQ}{b}$$

Potential at B is zero



11. above the surface

$$\frac{\Delta g}{g} = \frac{-2h}{R}$$

Below the surface

$$\frac{\Delta g}{g} = \frac{-x}{R}$$

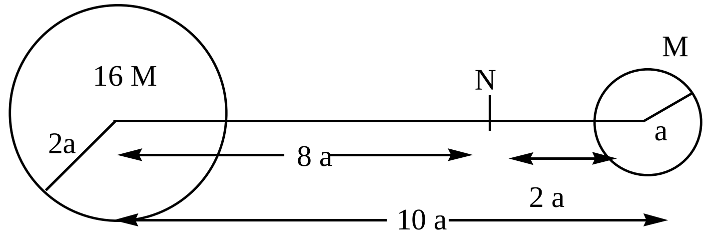
12. $(T.E)_P = (T.E)_{on\ surface}$

$$0 + \left[\frac{-GmM}{3R}\right] = \frac{1}{2}mv^2 + \left[\frac{-GmM}{R}\right]$$

$$\Rightarrow V = \sqrt{\frac{4GM}{3R}}$$

13. The velocity of 'm' from the bigger planet to reach smaller planet is such that it is just able to reach 'N'. Then due to influence of M it reaches to its surface. Null position from centre of bigger planet

$$r = \frac{10a}{\sqrt{\frac{M}{16m}} + 1} = 8a$$



Now, T. E of 'm' on the surface of bigger planet = T. E of 'm' at N

$$\Rightarrow \frac{1}{2}, V_{\min}^2 + \left[\frac{-Gm(16m)}{2a}\right] + \left[\frac{-GmM}{8a}\right] = 0 + \left[\frac{-Gm(16m)}{8a}\right] + \left[\frac{-GmM}{2a}\right]$$

$$\Rightarrow \ V_{\min} = \frac{3}{2}\sqrt{\frac{5GM}{a}}$$

14. Energy of the skylab in the first orbit is

$$-\frac{GMm}{2(2R)} = -\frac{GMm}{4R}$$

Total energy required to place the skylab into the orbit of radius 2R from the surface of earth is

$$-\frac{GMm}{4R}-\left(-\frac{GMm}{4R}\right)=\frac{3GMm}{4R}$$

$$=\frac{3gR^2m}{4R}=\frac{3}{4}mgR$$

Energy of the skylab in the second orbit = - (GMm)/ 6R.

Energy needed to shift the skylab from the first orbit to the second orbit is

$$-\frac{GMm}{4R}-\frac{GMm}{6R}=\frac{GMm}{R}\times\frac{2}{24}=\frac{mgR}{12}$$

15. At point P, we have $I_1-I_2=0$ (because the gravitational field inside a shell is zero). Hence $I_1=I_2$

16. $V=\sqrt{\frac{GM}{r}}$

Small object is ejecting with velocity $V_e=\sqrt{\frac{2GM}{r}}$

$\therefore$ It's $K.E=\frac{1}{2}mV_e^2=mv^2$

17. For $r\le R, R=kr$

$\Rightarrow \frac{mv^2}{r}=kr\Rightarrow v\propto r$

For r > R, $F=\frac{k}{r^2}\Rightarrow\frac{mv^2}{r}=\frac{k}{r^2}\Rightarrow v\propto\frac{1}{\sqrt{r}}$

18. Areal velocity $=\frac{dA}{dt}=\frac{L}{2m}=\frac{r^2\omega}{2}\Rightarrow\frac{dA}{dt}\alpha r^2\omega$

19. $x=\frac{d}{1+\sqrt{\frac{q_{big}}{q_{small}}}}=\frac{d}{1+\sqrt{\frac{4q}{q}}}=\frac{d}{3}$

$\frac{1}{4\pi\in_0}\frac{Qq}{x^2}=\frac{1}{4\pi\in_0}\frac{4q^2}{d^2}\Rightarrow Q=-\frac{4q}{9}\text{from 'q'}$

20. Points A and B lies on equipotential surface

21. $\frac{3g}{4}=g-R_e\omega^2$

$\omega=\frac{1}{2}\sqrt{\frac{g}{R_e}}$

$T=2\times2\pi\sqrt{\frac{R_e}{g}}=4\pi\sqrt{\frac{R_e}{g}}$

22. From conservation of energy

$\frac{-GM_em}{R_e}=\frac{1}{2}MV_1^2-\frac{GM_e}{R_e}\times\frac{11}{8}$

$v_1=\sqrt{\frac{3}{4}\frac{GM_e}{R_e}}$

$v_2=\sqrt{\frac{GM_e}{R_e}}$

$$\frac{v_1}{v_2}=\sqrt{\frac{3}{4}}=\frac{\sqrt{x+1}}{x} \quad \Rightarrow \quad x=2$$

23. $w_1=\frac{+GM_e m}{R_e}$

$w_2=\frac{3}{2}\frac{GM_e}{R_e}\times 2m$

$=3\ w_1$

$w_n=n\times\ w_1$

24. $g_p=\frac{GM_p}{R_p^2}=\frac{4}{3}G\pi R_p\rho_p$

$\Rightarrow \frac{g_p}{g_e}=\frac{R_p\rho_p}{R_e\rho_e}$

Also, $v_e=\sqrt{2gR}$

$\Rightarrow \frac{v_p}{v_e}=\sqrt{\frac{g_pR_p}{g_eR_e}}=\left(\frac{g_p}{g_e}\right)\sqrt{\frac{\rho_e}{\rho_p}}=\frac{\sqrt{6}}{11}\times\sqrt{\frac{3}{2}}$

$\Rightarrow v_p=3km/s$

25. $F=-K/r^2$ (negative sign is for attractive force)

Potential energy $U=-\int F\,dr=\int\frac{K}{r_2}dr=-\frac{K}{r}$

Conservation of energy gives (let at other extreme position r = b)

$K_1=U_1=K_2+U_2$

$\frac{1}{2}mv_1^2-\frac{K}{a}=\frac{1}{2}mv_2^2-\frac{K}{b}$ (i)

Where $v_1=\sqrt{\frac{K}{2ma}}$

Conservation of angular momentum gives

$mv_1a=mv_2b$

$v_2=\frac{a}{b}v_1=\frac{a}{b}\sqrt{\frac{K}{2ma}}$

Therefore, from Eq. (i)

$\Rightarrow \frac{1}{2}m\frac{K}{2ma}-\frac{K}{a}=\frac{1}{2}m\left(\frac{a}{b}\right)^2\frac{K}{2ma}-\frac{K}{b}$

$-\frac{3K}{4a}=\frac{aK}{4b^2}-\frac{K}{b}\Rightarrow b^2-\frac{4a}{3}b+\frac{a^3}{3}=0$

Hence, b = a/3 $\Rightarrow$ a/b = 3

26. $g=\frac{GM}{R^2}$

$g=\frac{\Delta g}{g}\times 100=-2\frac{\Delta R}{R}\times 100$

$=-2(-2)=+4\%$

27. $W=Fs\cos\theta=Eqs\cos\theta$

$$\therefore E = \frac{W}{q\,s\cos\theta} = \frac{3}{\left(2\times 2\cos 60^0\right)}$$

E = 1.5 N/C

28. $F_C = KR^{-5/2}$

$mR\omega^2 = kR^{-5/2}$

$\omega^2 \propto R^{-5/2-1}$

$\left(\frac{2\pi}{T}\right)^2 \propto R^{-7/2}$

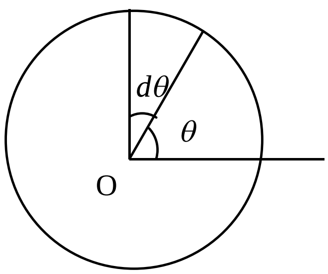
$\Rightarrow T^2 \propto R^{7/2}$

29. At stable equilibrium , $\theta_1 = 0^0$

Unstable equilibrium, $\theta_2 = 180^0$

$W = pE\left[\cos\theta_1 - \cos\theta_2\right] = PE\left[(1)-(-1)\right] = 2PE$

30. The charge on the infinitesimal elements of are that subtends an angle $d\theta$ at the center of the ring is



$dQ = \lambda R d\theta = \lambda_0 \cos\frac{\theta}{2} R d\theta$

Potential at the center of ring due to charges dQ is

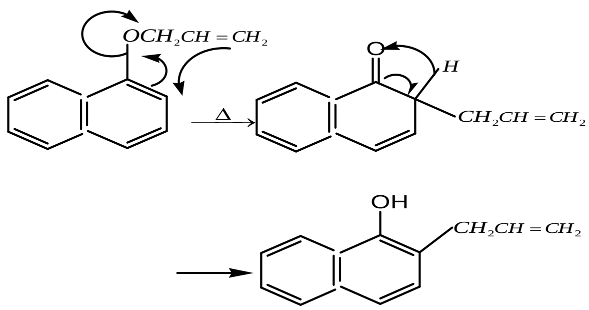
$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dQ}{R} = \frac{\lambda_0 \cos\frac{\theta}{2}.Rd\theta}{4\pi\varepsilon_0 R}$$

$$V = \int dV = \frac{\lambda_0}{4\pi\varepsilon_0}\int_0^{2\pi}\cos\frac{\theta}{2}d\theta$$

$$= \frac{\lambda_0}{4\pi\varepsilon_0}\left[\frac{\sin\frac{\theta}{2}}{\frac{1}{2}}\right]_0^{2\pi} = 0V$$

CHEMISTRY

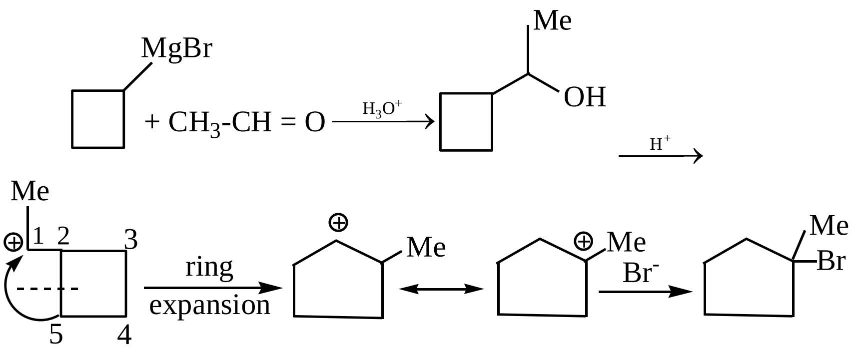
31.



32. Electron with drawing groups increase acidic nature of phenols

33. As the compound is soluble in $NaOH$, it is a phenol and cannot be anisole or benzyl alcohol .Out of O, m-cresols- only m-cresol gives tribromo derivative

34.



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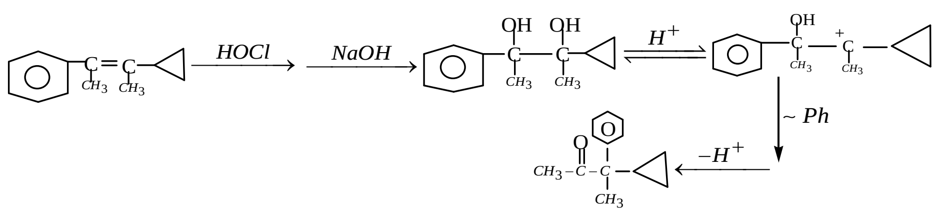


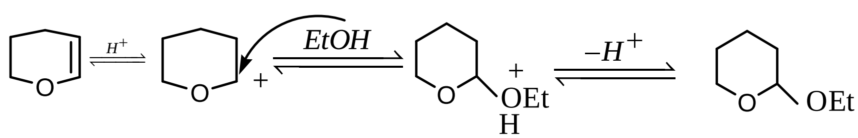
36. Williamson reaction is SN^2. A double bond in β position increases rate. Crowding on α carbon decreases rate. $S > Q > R > P$

37. Libermann's Reaction

38. Ring expansion (semi-Pinacol-Pinacolone re-arrangement)

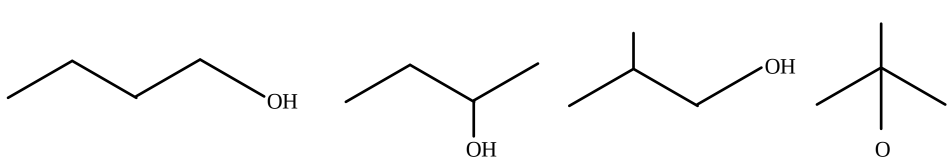
39. CONCEPTUAL

40.



41. More stable carbocation is formed

42.

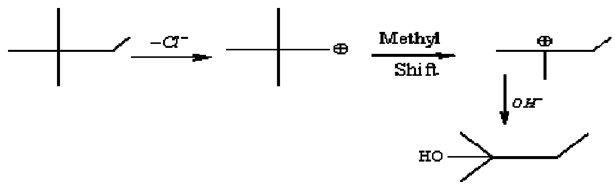


43. Conceptual

44. $CH_3CH_2OH \xrightarrow{H^+} CH_3CH_2\overset{\oplus}{O}H_2 \underset{Br^-}{\xrightarrow{-H_2O}} CH_3CH_2Br$

* H_2O is a good leaving group

45.



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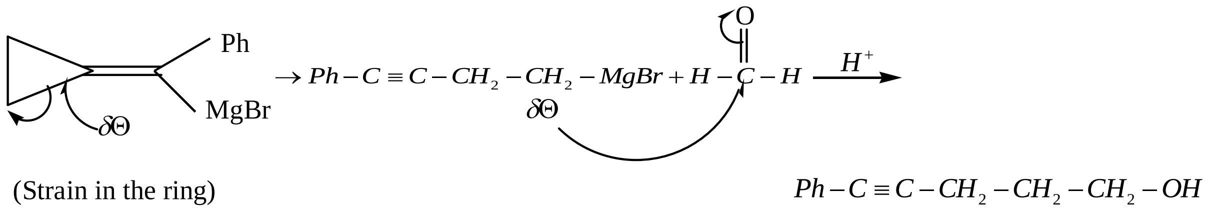

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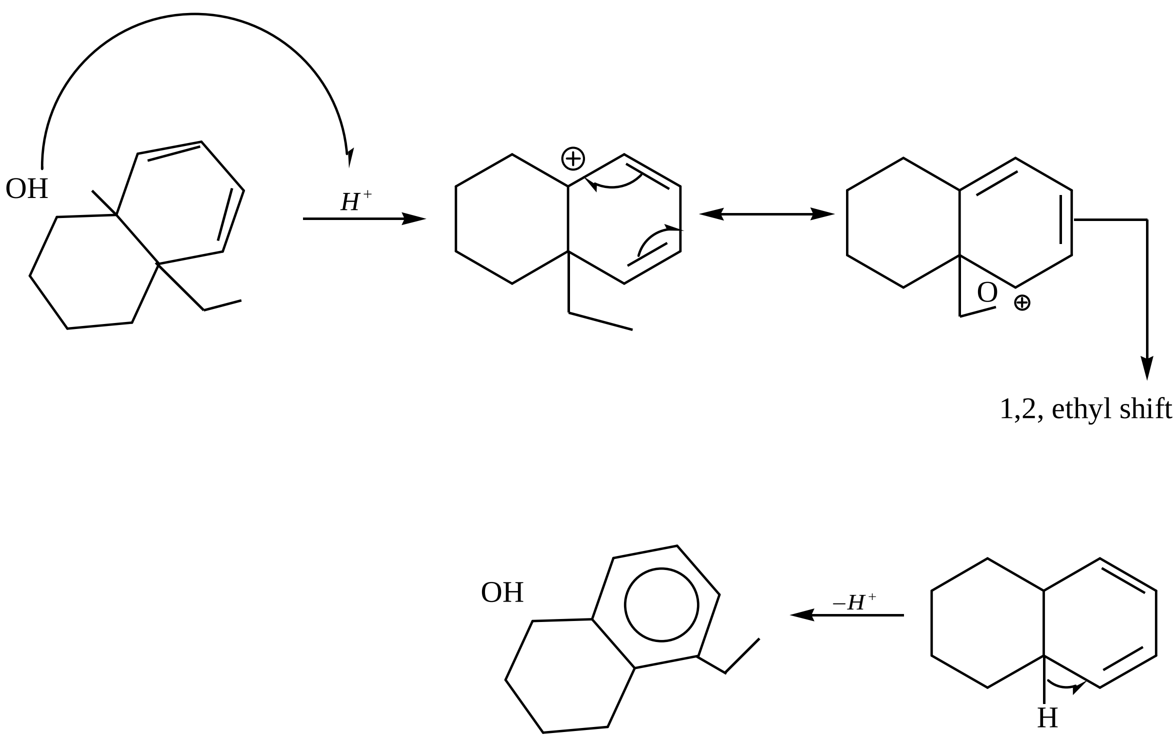
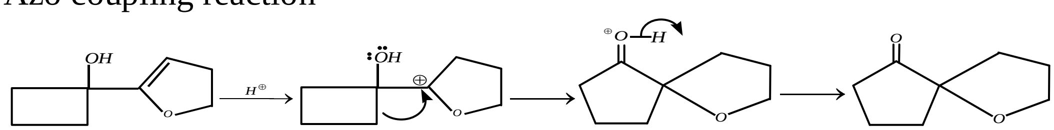


48. Azo coupling reaction

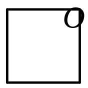
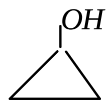
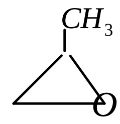
49.



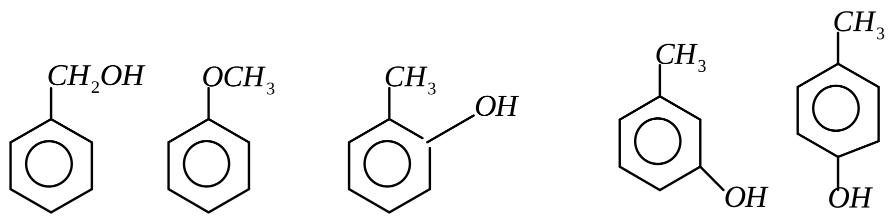
50. $$\%\text{Methoxy group} = \frac{31 \times \text{wt of AgI} \times 100}{\text{Mol.wt of AgI} \times \text{wt of ether}}$$

51. The number of isomers with the formula C_3H_6O including stereoisomers are 8 enolform not considered.

1. CH_3CH_2CHO 2. CH_3COCH_3 3. $CH_2=CH-\underset{|}{\underset{OH}{CH_2}}$ 4) $CH_2=CH-O-CH$

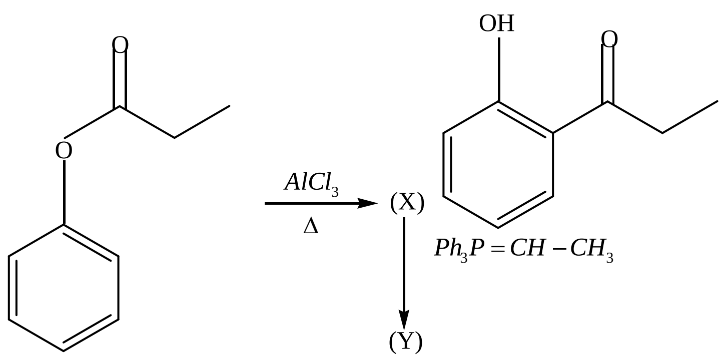
5.  6.  7., and 8 $(\pm)$ 

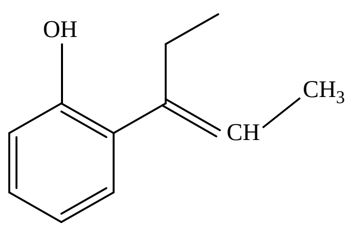
52. Number of isomers = 5



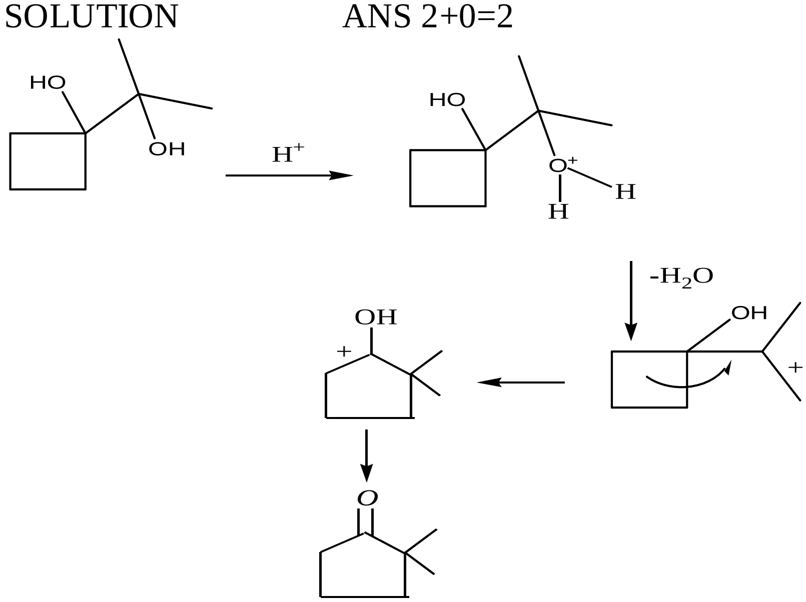
53. Number of moles of compound releases 3 moles of CH_4. Hence it has 3 – OH groups

54.

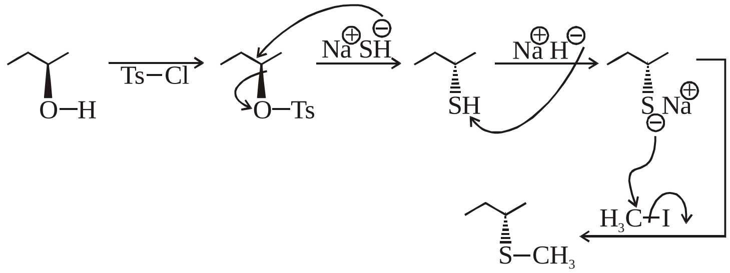




55. SOLUTION ANS 2+0=2



56. SOLUTION (6)



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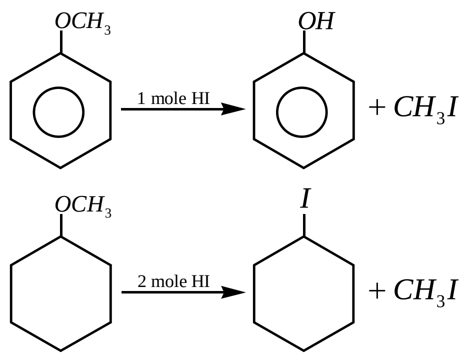


58. Dehydration

59. $C_2H_5OH + 4I_2 \rightarrow CI_3CHO + 5HI$

$4\left(O_3 + 2KI + H_2O \rightarrow 2KOH + I_2 + O_2\right)$

60. To reduce 1 mole CH_3COOH , 3 Hydride ions required. Totally 36 Hydride ions are required.

MATHEMATICS

61. Equation of the tangent is $y = 2x \pm \sqrt{4a^2 + b^2}$...........(1)

This is normal to the circle $x^2 + y^2 + 4x + 1 = 0$

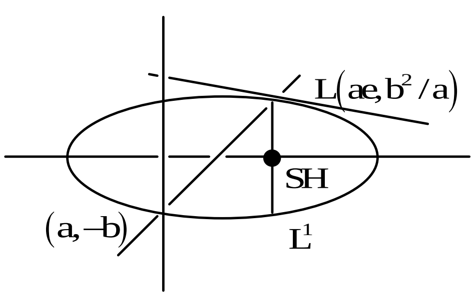
(1) passes through centre of the circle (-2,0)then

$0 = -4 + \sqrt{4a^2 + b^2} \Rightarrow 4a^2 + b^2 = 16$

$\text{A.M} \geq \text{G.M}$

$\frac{4a^2 + b^2}{2} \geq \sqrt{\left(4a^2\right).\left(b^2\right)} \Rightarrow 8 \geq 2ab \Rightarrow ab \leq 4$

62.



Equation of the normal at $L = \left(ae, \frac{b^2}{a}\right)$ is $\frac{a^2x}{ae} - \frac{b^2y}{b^2/a} = a^2 - b^2$

$\Rightarrow \frac{ax}{e} - ay = a^2e^2$

If it passes through $(0, -b)$, $o + ab = a^2e^2$

$\Rightarrow b = ae^2$

$\Rightarrow \frac{b^2}{a^2} = e^4 \Rightarrow 1 - e^2 = e^4$

$e^4 + e^2 - 1 = 0$

$e^2 = \frac{-1+\sqrt{1+4}}{2} \Rightarrow e^2 = \frac{-1+\sqrt{5}}{2} = 2\left(\frac{\sqrt{5}-1}{4}\right) = 2\left(\cos 72^0\right)$

63. Major axis on y-axis

$f\left(a^2-5\right) > f\left(4a\right)$

$a^2 - 5 < 4a \quad (\because f \text{ is } \text{ decreasing})$

$a^2 - 4a - 5 < 0$

$(a-5)(a+1) < 0 \Rightarrow -1 < a < 5$

64. At $t = \frac{3}{2}, \ a = \left[\frac{9}{4} - \frac{9}{2} + 4\right], \ b = \left[3 + \frac{15}{2}\right] \Rightarrow a < b$ or $b > a$

L.L.R $= \frac{2a^2}{b} = \frac{2(1)^2}{10} = \frac{1}{5}$

65. Equation of tangent to the parabola $y^2 = 32x$ in $y = mx + \frac{8}{m}.....(1)$

(1) in also touches the hyperbola $x^2 - y^2 = \frac{8}{9}$

i.e $\left(\frac{8}{m}\right)^2 = \frac{8}{9} \times m^2 - \frac{8}{9}$

$\frac{1}{m^2} = \frac{m^2}{9} - \frac{1}{9} \Rightarrow m^4 - m^2 - 72 = 0$

$\Rightarrow \left(m^2 - 9\right)\left(m^2 + 8\right) = 0$

$m^2 = 9 \Rightarrow m = \pm 3$

Eq of the tangent $y = 3x + \frac{8}{3} \Rightarrow 9x - 3y + 8 = 0$

66. H = $x^2 + 3xy + 2y^2 + 2x + 3y = 0$ centre $\left(\frac{\partial}{\partial x} = 0, \frac{\partial}{\partial y} = 0\right) \quad c = (-1, 0)$

Let pair of a symptiotes in $x^2 + 3xy + 2y^2 + 2x + 3y + \lambda = 0$ ----------(1)

Centre lies on (1) , $\lambda = 1$

$\therefore A = x^2 + 3xy + 2y^2 + 2x + 3y + 1 = 0$

Since H + c = 2A

$C = 2A - H$

$\therefore C = x^2 + 3xy + 2y^2 + 2x + 3y + 2 = 0$

67. Equation of normal parabola $y^2 = 4ax$ is $y = mx - 2am - am^3$ which is a tangent of

$x^2 - y^2 = a^2$

$c^2 = a^2m^2 - b^2$

$\Rightarrow \left(-2am - am^3\right)^2 = a^2m^2 - a^2$

$4m^2 + m^6 + 4m^4 = x^2 - 1$

$\Rightarrow m^6 + 4m^4 + 3m^2 + 1 = 0$

68. $x^2 - y^2\sec^2\alpha = 5 \Rightarrow \frac{x^2}{5} - \frac{y^2}{5\cos^2\alpha} = 1,$

$e_1{}^2 = \frac{a^2+b^2}{a^2} = \frac{5+5\cos^2\alpha}{5} = 1+\cos^2\alpha$

$E: \frac{x^2}{25\cos^2\alpha} + \frac{y^2}{25} = 1, e_2^2 = \frac{25-25\cos^2\alpha}{25} = 1-\cos^2\alpha = \sin^2\alpha$

Since $e_1 = \sqrt{3}\ e_2 \Rightarrow e_1^2 = 3e_2^2$

$\Rightarrow 1+\cos^2\alpha = 3\sin^2\alpha$

$\Rightarrow 2 = 4\sin^2\alpha \Rightarrow \sin\alpha = \frac{1}{\sqrt{2}} = \sin\frac{\pi}{4} \Rightarrow \alpha = \frac{\pi}{4}$

69. Locus of the mid points of chord is $s_1 = s_{11} \Rightarrow xx_1 + yy_1 = x_1^2 + y_1^2$

This chord tangents the hyperbola

$a^2l^2 - b^2m^2 = n^2 \Rightarrow 9x_1^2 - 16y_1^2 = \left(x_1^2+y_1^2\right)^2$

Locus of (x_1, y_1) is $\left(x^2+y^2\right)^2 - 9x^2 + 16y^2 = 0$

70. Any point on the ellipse $p\left(\sqrt{6}\cos\theta, \sqrt{2}\sin\theta\right)$,where $0 \le \theta < 2\pi$,centre(0,0) $CP = \sqrt{5}$

$6\cos^2\theta + 2\sin^2\theta = 5 \Rightarrow \sin^2\theta = \frac{1}{4} \Rightarrow \sin\theta = \pm\frac{1}{2}$

$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\quad \text{sum} = 4\pi$

71. $x = k^2+1, y = 2k$ and $x = 2t, y = \frac{2}{t}$

$x - 1 = \frac{k^2}{4}$ $\qquad xy = 4$ ....(2)

$y^2 = 4(x-1)$....(1)

From (1) and (2) $(x, y) = (2, 2)$

72. Equation $\left(x - \frac{1}{13}\right)^2 + \left(y - \frac{2}{13}\right)^2 = a\frac{(5x+12y-1)^2}{\left(\sqrt{25+144}\right)^2}$

$\sqrt{\left(x-\frac{1}{13}\right)^2 + \left(y-\frac{2}{13}\right)^2} = \sqrt{a}\,\frac{(5x+12y-1)}{\sqrt{169}}$

SP=e.(pm)

$0 < a < 1$

$a \in (0,1)$

73. Equation of on ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ......(1)

(1) meets $\frac{x}{7} + \frac{y}{2} = 1$ on the axis of x,$(y = 0)$

$\frac{x}{7} + 0 = 1$ x = 7 $\Rightarrow$ a = 7

1) meets $\frac{x}{3} - \frac{y}{5} = 1$ on axis of y,(x = 0)

$0 - \frac{y}{5} = 1 \Rightarrow y = -5 \Rightarrow b = -5$

$b^2 = a^2\left(1-e^2\right)$

$x = 49\left(1-e^2\right)$

$49e^2 = 24$

$e^2 = \frac{24}{49} \Rightarrow e = \frac{2\sqrt{6}}{7}$

74. $p = \left(\frac{\pi}{6}\right) = p\left(\sec\frac{\pi}{6}, b\tan\frac{\pi}{6}\right) = p\left(\frac{2a}{\sqrt{3}}, \frac{b}{\sqrt{3}}\right)$

Eq. Of the tangent at p is $\frac{2x}{\sqrt{3}a} - \frac{y}{\sqrt{3}b} = 1$

x-intercept=$\frac{\sqrt{3}a}{2}$, y-intercept=$-\sqrt{3}b$

Area of triangle formed by the tangent with the coordinate axis is

$\frac{1}{2}\left|\frac{\sqrt{3}a}{2} x \sqrt{3}b\right| = 3a^2$

$\frac{b}{a} = 4 \Rightarrow \frac{a}{b} = \frac{1}{4}$

Eccentricity of the conjugate hyperbola in $\sqrt{1+\frac{1}{16}} = \frac{\sqrt{17}}{4}$

75. Let $s = (4,2), s^1 = (2,2)$

$SS^1 = 2ae$ and $p(1,2) sp + s^1p = 2a$

$e = \frac{SS^1}{sp + s^1p} = \frac{2}{2\sqrt{2}+2} = \frac{1}{\sqrt{2}+1} = \sqrt{2}-1 = \tan\frac{\pi}{8}$

76. Eq. of the chord joining α and β is $\frac{x}{a}\cos\left(\frac{\alpha-\beta}{2}\right) - \frac{4}{b}\sin\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha+\beta}{2}\right)$

Since $\alpha+\beta = 3\pi$, then $\frac{x}{a}\cos\left(\frac{\alpha-\beta}{2}\right) + \frac{y}{b} = 0$

If it passes through the centre (0,0)

77. Given the B eq. is $xdy + ydx = 0$

$d(xy) = 0 \Rightarrow xy = c$,which passes through (2,8) is C=16

Eq. of the length $xy = 16 = 4^2$

Length of the latusrectum=$2\sqrt{2}(4) = 8\sqrt{2}$

78. Tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$is $y = mx \pm \sqrt{a^2m^2 - b^2}$.............(1)

Given the tangent is $y = \alpha x + \beta$.............(2)

From (1) and (2) $m = \alpha,\quad a^2m^2 - b^2 = \beta^2$

$a^2\alpha^2 - \beta^2 = b^2$

Locus of $(p/\alpha,\beta)$ is $ax^2 - y^2 = b^2$, represents a hyperbola

79. Here $a^2 = \cos^2\alpha, b^2 = \sin^2\alpha$

Since $b^2 = a^2\left(e^2-1\right)$

$\sin^2\alpha = \cos^2\alpha\left(e^2-1\right) \Rightarrow \sin^2\alpha + \cos^2\alpha = e^2\cos^2\alpha$

$\Rightarrow 1=\dfrac{e^2}{\sec^2\alpha}\Rightarrow e^2=\sec^2\alpha\Rightarrow e=\sec\alpha$

Now $ae=\cos\alpha.\sec\alpha=1$

Foci are $|\pm ae,0|=(\pm 1,0)$

Abscissa of foci remain constant

80. Any tangent to the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ at $p(\theta)$ is $\dfrac{x\cos\theta}{a}+\dfrac{y\sin\theta}{b}=1$

Tangent meets the coordinate axis at $A=(a\sec\theta,0),B=(0,b\operatorname{cosec}\theta)$

Area= $\dfrac{1}{2}|(a\sec\theta)||b\operatorname{cosec}\theta|=\dfrac{ab}{\sin 2\theta}$, (Area to be min, $\sin 2\theta$ should max)

81. Eq. of the tangent with slope 3 to the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{1}=1$ in $y=3x\pm\sqrt{9a^2+1}$

If it passes through $(-2,0)$ $0=-6\pm\sqrt{9a^2+1}$

$9a^2+1=36$

$a^2=\dfrac{35}{9}=3.88$ $\left[a^2\right]=3$

82. Equation of the normal is

$\dfrac{5x}{\cos\theta}-\dfrac{y}{\sin\theta}=25-1$

$(5\sec\theta)x-(\operatorname{cosec}\theta)y=24$

$P=\dfrac{24}{\sqrt{25\sec^2\theta+\operatorname{cosec}^2\theta}}$

$=\dfrac{24}{\sqrt{36+(5\tan\theta-\cot\theta)^2}}$

$\therefore$ maximum value of $r=\dfrac{24}{6}=4$

83. Let $P=(4\cos\theta,2\sin\theta),Q=\left(4\cos\left(\theta+\dfrac{\pi}{6}\right),2\sin\left(\theta+\dfrac{\pi}{6}\right)\right)$

Area of are OPQ = $\dfrac{1}{2}\begin{vmatrix}4\cos\theta & 2\sin\theta & 1\\ 4\cos\left(\theta+\dfrac{\pi}{6}\right) & 2\sin(\theta+\pi/6) & 1\\ 0 & 0 & 1\end{vmatrix}$

$4\left[\left(\cos\theta.\sin\left(\theta+\dfrac{\pi}{6}\right)\right)-\cos\left(\theta+\dfrac{\pi}{6}\right).\sin\theta\right]$

$=4\sin\left(\theta+\dfrac{\pi}{6}-\theta\right)=4.\dfrac{1}{2}=2\text{square units}$

84. Equation of the hyperbola is $\dfrac{x^2}{2}-\dfrac{y^2}{3}=1$

Equation of the tangent $y=mx\pm\sqrt{a^2m^2-b^2}$

Tangent from the point (h,k) is $(k-hm)^2=2m^2-3$

$k^2-2hkm+h^2m^2=2m^2-3$

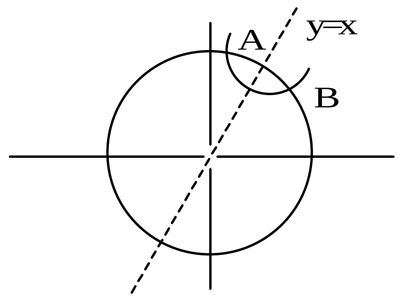
$m^2(h^2-2)-2hkm+k^2+3=0$

Since $m_1m_2 = \frac{k^2 = 3}{h^2 - 2} = 2\tan\alpha\tan\beta$

$k^2 + 3 = 2h^2 - 4 \Rightarrow 2h^2 - k^2 = 7$

85. Let $A = \left(t, \frac{1}{t}\right), B = \left(\frac{1}{t}, t\right)$

Since $AB = \sqrt{14} \Rightarrow AB^2 = 14$



$\left(t - \frac{1}{t}\right)^2 + \left(\frac{1}{t} - t\right)^2 = 14$

$2\left(t^2 + \frac{1}{t^2}\right) = 14 + 4 = 18$

$t^2 + \frac{1}{t^2} = 9 \Rightarrow (2r)^2 = 9 \Rightarrow |2r| = 3 \Rightarrow |r| = 1.5$

86. Given $2.a.e = 10 \Rightarrow a^2e^2 = 25 \Rightarrow a^2 + b^2 = 25$............(1)

Since $(2, \sqrt{3})$ lies on the director circle $x^2 + y^2 = a^2 - b^2$

$7 = a^2 - b^2$...........(2)

From (1) and (2) $a^2 = 16, b^2 = 9$

$\left|\frac{b}{a}\right| = \left|\frac{3}{4}\right| = \frac{3}{4} = 0.75$

87. $\frac{x^2}{16} + \frac{y^2}{b^2} = 1,$ ....(1)

$x^2 + y^2 = 4b,\ b > 4,$ ....(2)

$y^2 = 3x^2$ ....(3)

From (2) and (3) $4x^2 = 4b \Rightarrow x^2 = b$, $y^2 = 3b$

From (1) $\frac{b}{16} + \frac{3b}{b^2} = 1 \Rightarrow b^2 + 48 = 16b$

$b^2 - 16b + 48 = 0$

$(b - 12)(b - 4) = 0$

$b = 12$

88. Eq. Of the circle $x^2 + y^2 = a^2$...........(1)

Hyperbola $xy = c^2$.........(2)

From (1) and (2) $x^2 + \frac{c^4}{x^2} = a^2 \Rightarrow \frac{c^4}{y^2} + y^2 = a^2$

$x^4 - a^2x^2 + c^4 = 0 \Rightarrow y^4 - a^2y^2 + c^4 = 0$

$\sum x_1 = 0\ \pi(x_1) = c^4 \quad \sum y_1 = 0, \pi(y_1) = c^4$

$\sum x_1 + \sum y_1 - \pi(x_1) + \pi(y_1) = 0$

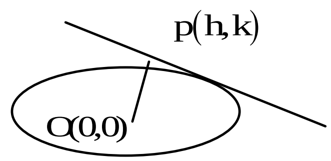
89. Given hyperbola is $x^2 - y^2 = a^2, \quad e = \sqrt{2} \qquad \left(x_1^2 - y_1^2 = a^2\right)$

Since $(PF_1)(PF_2) = (ex_1 - a)(ex_1 + a) = e^2x_1^2 - a^2 = 2x_1^2 - a^2$

$= x_1^2 + y_1^2 = OP^2$

$(PF_1)(PF_2) = 1.(OP)^2 \Rightarrow \lambda = 1$

90. Slope of $OP = \frac{k}{h}$



Slope of the tangent $= -\frac{h}{k}$

Eq. Of the tangent in $y - k = \frac{-h}{k}(x - h) \Rightarrow y = -\frac{h}{k}x + \frac{h^2 + k^2}{k}$

Since $c^2 = a^2m^2 + b^2 \Rightarrow \left(\frac{h^2 + k^2}{k}\right) = 40\left(-\frac{h}{k}\right)^2 + 10$

The locus of $(-h, k)$ is $\left(x^2 + y^2\right)^2 = 40h^2 + 10k^2 \qquad p = 40, q = 10$